

MEDIATEK

MT8676 Hypervisor AI User Manual

Version: 1.0
Release date: 2024-11-13

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Version History

Version	Date	Author	Description
1.0	2024-11-13	Wmeng.Wu	Official release

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1 AI

1.1 Overview

This section describes the functions related to MT8676 AI/VPU.

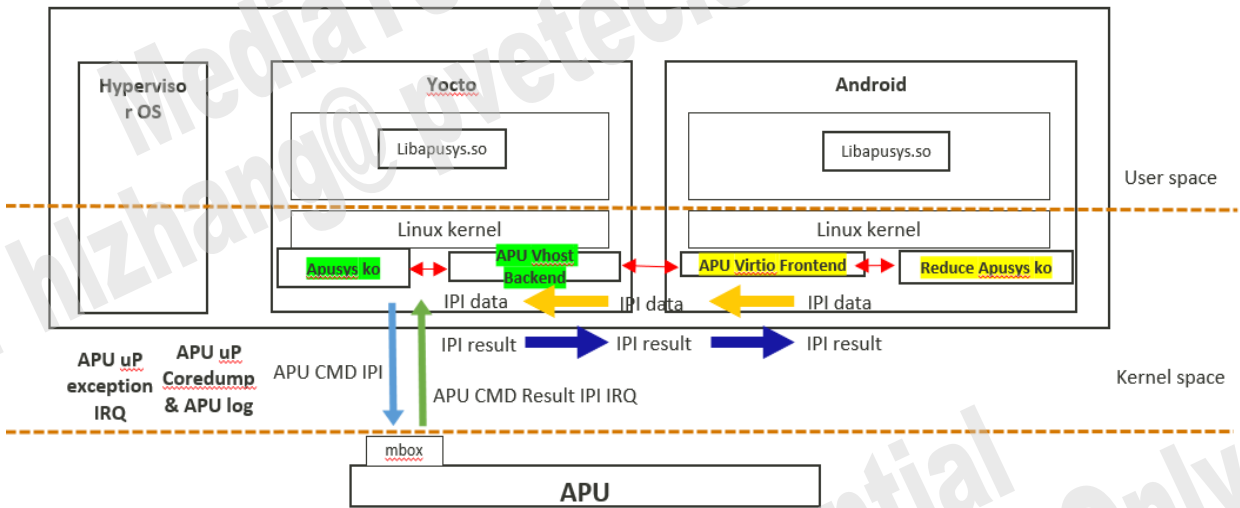
1.1.1 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
AI	Artificial Intelligence
MDLA	MediaTek Deep Learning Accelerator
MVPU	MediaTek Vision Processing Unit
NeuroPilot	MediaTek’s Ecosystem for AI Development

1.2 Architecture/Process Overview

1.2.1 APU Virtualization



Improved compatibility: After APU virtualization, using the NeuroPilot approach for the User level, which remains consistent with a single system.

1.2.2 AI Architecture

NeuroPilot is a set of software tools and APIs for developing efficient artificial intelligence applications on the MediaTek platform; it is the core of the MediaTek artificial intelligence ecosystem. “Edge AI” is supported by NeuroPilot, which

means executing AI on local devices rather than remotely on servers. This can result in faster AI tasks while also protecting data and privacy.

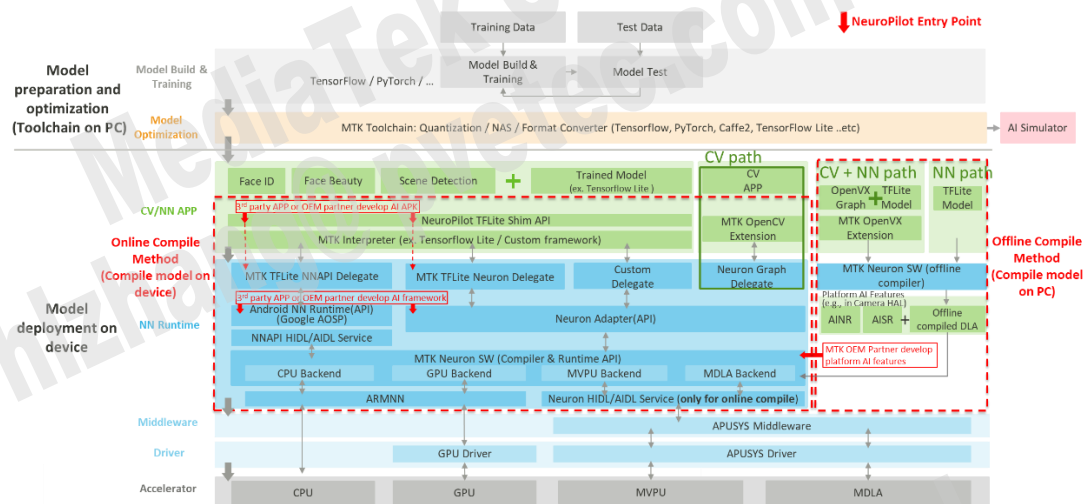


Figure 1-1. AI architecture

Currently, the MT8676 software stack is shown in Figure 1-1, and it mainly includes the following layers:

- **CV/NN Application Layer:** This layer contains user-written code for running AI applications. The layer also includes the MediaTek interpreter, which is an Android TensorFlow Lite interpreter optimized for MediaTek-NPU, and the TFLite Shim API, a wrapper layer built on top of the MediaTek interpreter aimed at simplifying API calls.
- **NN runtime layer:** This layer contains the runtime libraries that provide NN acceleration, including NNAPI and MediaTek Neural Compiler/Runtime.
- **Middleware layer:** This layer allows dynamic control of the MediaTek AI computing core and provides service quality control for NN workloads.
- **Driver Layer:** This layer provides drivers for the specialized MediaTek AI computing core.

1.2.3 NeuroPilot Development Guide

To access NeuroPilot Online Document, customers are required to apply for an account first. This account can then be used to access MediaTek online document website. The website provides various development materials including development-related data, conversion tools, SDKs, SampleCode, as well as information on the support and limitations of model OP for each target. Customers seeking access to the online document can reach out to CPM for assistance with the application process.

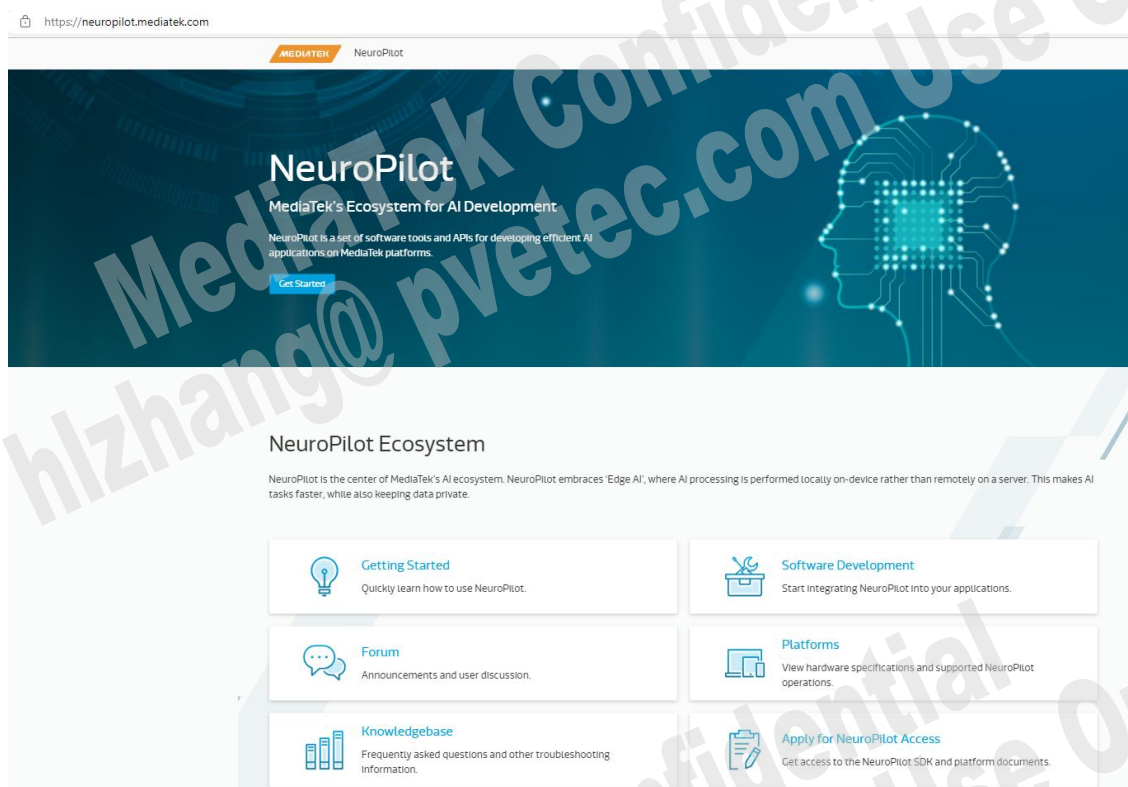


Figure 1-2. NeuroPilot online document

After applying for access to NeuroPilot, go to <https://neuropilot.MediaTek.com/> -> Software Development -> log in to your account -> under "NeuroPilot SDK & Document", select the target NeuroPilot {version} -> Latest Version NeuroPilot Online Doc -> 2. Getting Started Guide -> 2.2. NeuroPilot Workflow -> 2.2.3.1. Android Development.

1.3 Configuration/Customization Guideline

1.3.1 Explanation of NeuroPilot Debug Commands

If customers encounter an apusys error issue, please first turn on the following log switches, reproduce the problem, and then provide MediaTek with the complete machine log and the apusys_rv_xfile (used to decode apusys_log) on the test machine. The log options and retrieval paths are as follows:

```
Enable NNAPI AOSP log : adb shell "setprop debug.nn.vlog 1"
Enable TFlite log : adb shell setprop debug.mtk_tflite.vlog true
Enable Execution plan : adb shell setprop debug.neuron.runtime.ShowExecPlan true
Enable ShowQoSInfo : adb shell setprop debug.neuron.runtime.ShowQoSInfo true
Enable Kernel Log : adb shell "echo 15 > /sys/class/misc/apusys/log/klog"
Enable uPLog : adb shell "echo 5 > /proc/apusys_logger/log"
Enable User Log : adb shell setprop debug.apusys.loglevel 15
Enable apusys_rv_xfile: adb pull /proc/apusys_rv/apusys_rv_xfile (used for decoding apusys_log)
```

After turning on these required switches, a service restart is needed for the changes to take effect:

- adb shell stop neuralnetworks_hal_service_mtk_neuron
- adb shell start neuralnetworks_hal_service_mtk_neuron

1.3.2 The APU Trace Tool Captures the Trace

1.3.2.1 Use the Trace tool to Obtain the Path

APU Systrace is a tool used for analyzing and debugging the operational status of basic computing units (such as EDMA, MDLA, MVPUP) on the AI Processing Unit (APU) of MediaTek platforms. It helps developers quickly analyze the performance of model algorithms running on the APU and debug issues.

To get the APU trace tool, go to <https://neuropilot.MediaTek.com/> -> Software Development -> log in -> select the target NeuroPilot {version} under "NeuroPilot SDK & Document" -> Latest Version NeuroPilot Online Doc -> Downloads to download APU Systrace Tool.

1.3.2.2 How to Use APU Trace Tool

To record APU trace, execute 02-trace_start_all.bat -> run the test program -> 02-trace_stop.bat.

- 02-trace_start_all.bat - start recording trace
- 02-trace_stop.bat - stop tracing and pull trace files

Running the two scripts below is optional. Enabling them will result in the system trace containing APU middleware and neuron trace.

- 08-mdw_trace_enable.bat - To obtain tracing information from the APU middleware.
- 08-neuron_rt_trace_enable.bat - To obtain tracing information from the neuron runtime.

Three files will be generated in the end, choose to view as needed (Please use <https://ui.perfetto.dev/> to open the trace.)

- apusys.trace - apusys trace only
- System trace - system trace only
- combine.trace - apusys + system trace

1.3.2.3 Analysis Example of APU Trace

The trace files obtained by capturing and parsing with MediaTek APU Systrace usually have the following three:

- apusys.trace (Only includes the status of tasks on various devices such as MDLA/MVPUP during APU operation and information on APU Frequency, DRAM access, TCM access, etc.)
- system.trace (Regular system trace only, including CPU information and information about other processes/threads in the system)
- combine.trace (apusys + system trace)

In actual scenarios, there is often a need to clarify the debugging requirements for the threads running APU in the current system. The following SOP can be referred to:

- 1. Open the apusys.trace file using Perfetto UI, find the Tasks block running on the MDLA or MVPU (Main AI Compute Unit) Core, and you can obtain the following information:
 - Does the corresponding Task run in SMP multi-core parallel mode? If running in multi-core MDLA mode, there will be multiple Tasks with the same color/pid as shown in [Figure 1-3](#):



Figure 1-3. APU HW status

- The DRAM/TCM occupancy situation corresponding to the Task
- The operating frequency of the MDLA/MVPU Cores during the corresponding Task execution period
- You can also check whether the AI algorithm is executed periodically by running Tasks with the same pid on MDLA Core.

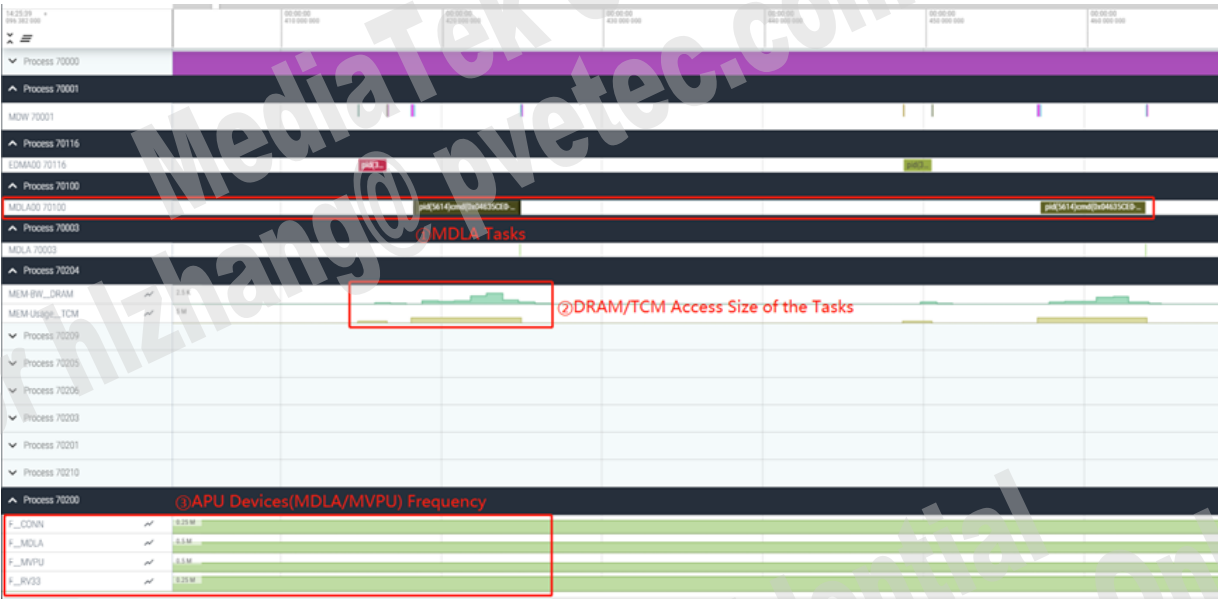


Figure 1-4. APU frequency status

1.4 Information about the Operators Supported by Specific MediaTek Platform's NPU

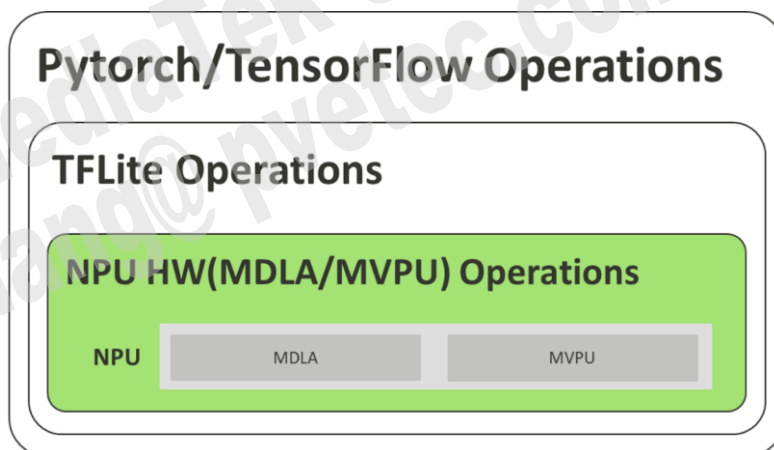


Figure 1-5. Relationship of supported operators

As shown in [Figure 1-5](#), the operators supported by the MediaTek platform NPU are divided into three levels from small to large:

- **PyTorch/TensorFlow Ops -> TFLite Ops:** By using the mtk_converter tool, the original Ops in .pt or .pb models are converted to TFLite Ops. This mapping process will undergo initial Ops filtering to block Ops not supported by the platform's NPU (HW). For a list of PyTorch/TensorFlow Ops that can be recognized and converted to TFLite Ops by the converter tool, please refer to the Online Document: Developer Tools -> Model Development -> Converter -> Converter Tool Supported Operators.
- **TFLite Ops -> NPU HW(MDLA/MVPU) Operations:** By using the ncc-tflite (compiler) in neuronsdk, the converted TFLite model is compiled into a dla file. During this mapping process, the Specification (Restrictions) in the NPU HW (MDLA/MVPU) Operations Guidelines will be referenced to check the detailed parameters of each op in TFLite. For information on which TFLite Ops can be recognized and compiled into dla files by the ncc-tflite tool, please refer to Supported Operations.
- **NPU HW (MDLA/MVPU) Operations:** Ops that can truly run on NPU (MDLA or MVPU).

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